

★ WEEK 7 · PART 1 of 3

Cointegration: The Long-Run Anchor

Two Wanderers That Can't Drift Too Far Apart

The big idea · the equilibrium spread · the cure for spurious regression

Individually, two price series can wander without bound. Yet a special combination of them may stay put — a long-run equilibrium. That tethering is cointegration, and it anchors pairs trading.

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PART 1 · COINTEGRATION

Tethered, Not Tied Down

The idea that each series wanders but their combination does not, the formal definition, the intuition, and how cointegration finally cures the spurious-regression trap of Week 2.

The Problem Granger Solved — Twice — Part 1

From exposing a trap to building its cure

In 1974, Clive Granger helped expose the spurious-regression trap (Week 2): unrelated trending series can look strongly related. Having revealed the danger, he then set out to fix it.

The trap he exposed

- Regressing one random walk on another gives a high R-squared.
- The relationship is a statistical illusion.
- It warned a generation off naive regressions of levels.

The question it raised

- If levels usually can't be regressed, are they ever safe to?
- Is there a case where a relationship in levels is REAL?
- Answering that led to cointegration.

The Problem Granger Solved — Twice — Part 2

Engle and Granger, 1987, and the lesson

Engle and Granger's 1987 paper defined cointegration and its error-correction counterpart — the exception where a levels relationship is genuine. It helped earn the 2003 Nobel.

What they showed

- Some $I(1)$ series share a genuine long-run equilibrium.
- A special combination of them IS stationary.
- Then regressing their levels is valid, not spurious.

The lesson for us

- Not all levels regressions are traps — cointegrated ones are real.
- The stationary combination is an economic anchor.
- That anchor is exactly what a pairs trade exploits.

From Short-Run Dynamics to a Long-Run Link

Where Week 6 left off

1

VAR needed stationary series

Week 6 modeled joint short-run dynamics — but a standard VAR requires differencing trending series first, which can discard their long-run relationship.

2

Some pairs share a long-run equilibrium

Two non-stationary prices may nonetheless move together forever, tied by economics. Differencing both would throw that tie away.

3

Cointegration models the tie directly

It captures the stable relationship in LEVELS, and error correction (Part 3) describes how deviations are pulled back. This is the pairs-trading engine.

Part 1 in one line: two series can each wander forever, yet their combination stays anchored.

The Big Idea of Cointegration

Each series wanders; their combination does not

Take two non-stationary series that each drift without bound. If some fixed linear combination of them is stationary, they are cointegrated — bound together by a long-run equilibrium.

Individually

- Each series is $I(1)$ — a wandering unit-root process.
- Neither has a mean it returns to.
- On its own, each looks unpredictable in level.

Together

- A specific combination cancels the shared wandering.
- What remains is stationary — it mean-reverts.
- That stationary combination is the equilibrium.

Cointegration, Formally

One line that captures the whole idea

$$X_t, Y_t \sim I(1) \quad \text{but} \quad z_t = Y_t - \beta X_t \sim I(0) \Rightarrow \text{cointegrated}$$

Reading it

- X and Y are each $I(1)$ — non-stationary.
- But the spread $z = Y - \beta X$ is $I(0)$ — stationary.
- That is the definition of cointegration.

The key word

- It only works for one special value of β .
- For any other β , the combination still wanders.
- Finding that β is the heart of the whole method.

The Integration Bookkeeping

Normally impossible — except when cointegrated

$$\underbrace{I(1)}_{Y_t} - \beta \underbrace{I(1)}_{X_t} = \underbrace{I(0)}_{\text{stationary spread}} \quad \text{only for the special } \beta$$

The usual rule

- Combining two $I(1)$ series generally gives another $I(1)$.
- The wandering does not cancel — it compounds.
- So a stationary combination should be impossible.

The exception

- Unless the two share the SAME stochastic trend.
- Then the right β subtracts one trend from the other.
- The trends cancel, leaving a stationary remainder.

The Drunk and Her Dog

The metaphor that makes cointegration click

A classic image: a drunk walks home randomly; her dog wanders randomly too. Each path is a random walk. But a leash keeps them close — so the DISTANCE between them is stationary.

The two random walks

- The drunk's position wanders unpredictably.
- The dog's position wanders unpredictably too.
- Neither one, alone, returns to any fixed spot.

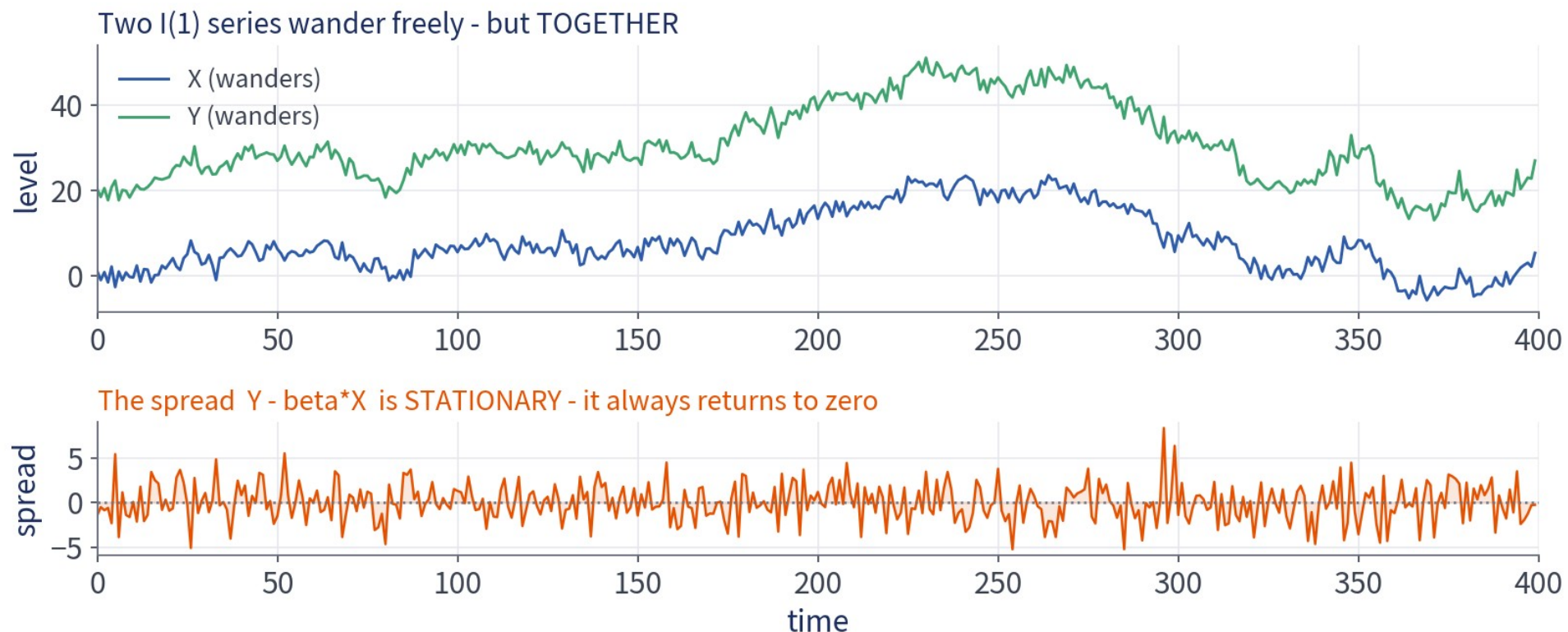
The leash = cointegration

- The leash ties them: they cannot drift far apart.
- Their separation mean-reverts around the leash length.
- That bounded gap is the stationary spread.

Tethered, Not Tied Down

Both series wander freely, yet always together (top). Their spread $Y - \beta X$ is stationary — it repeatedly returns to zero (bottom). That returning spread is the anchor.

Tethered, not tied down: cointegration



The Equilibrium Spread

The stationary combination, and what it means

What the spread is

- The stationary combination $z = Y - \beta X$.
- It measures the deviation from the long-run equilibrium.
- Zero means the pair is 'fairly' priced relative to each other.

Why it is powerful

- A positive spread $\rightarrow$ Y is rich, X is cheap (relatively).
- A negative spread $\rightarrow$ the reverse.
- Because it mean-reverts, extremes are trading opportunities.

The Spurious-Regression Cure

Week 2's villain, finally defeated

Spurious regression warned us never to regress one trending series on another. Cointegration is the precise exception: when the residual is stationary, the regression is genuine.

The spurious case (Week 2)

- Two unrelated random walks give a high but meaningless R-squared.
- The regression residual itself wanders — it is $I(1)$.
- The relationship is a mirage.

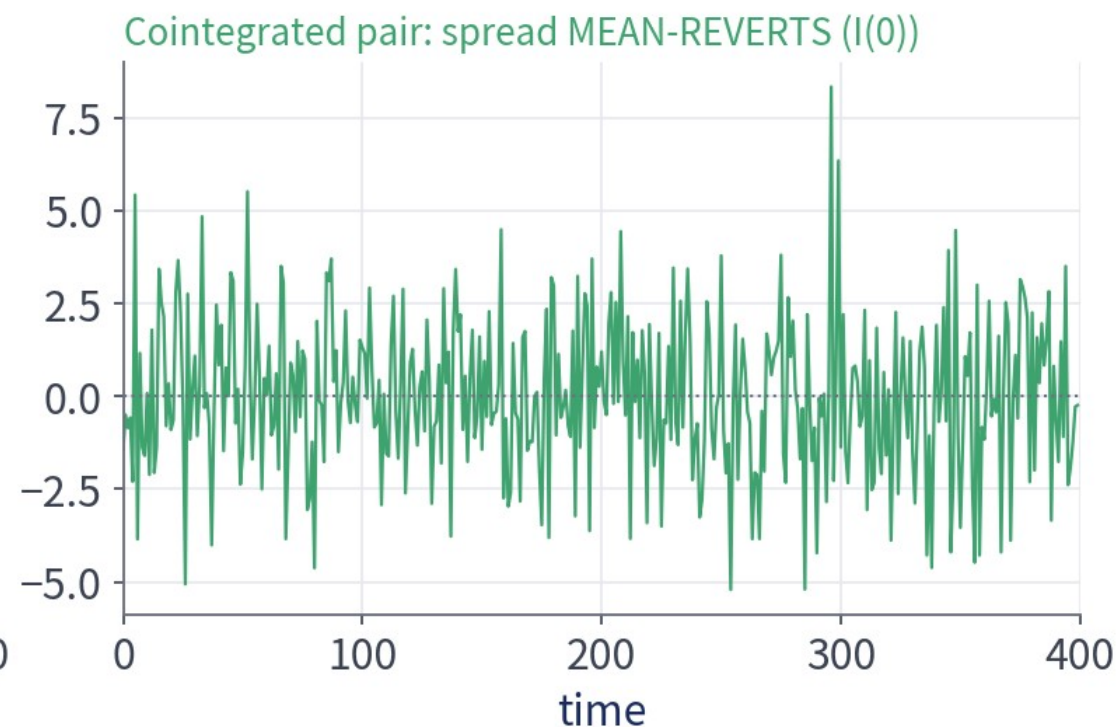
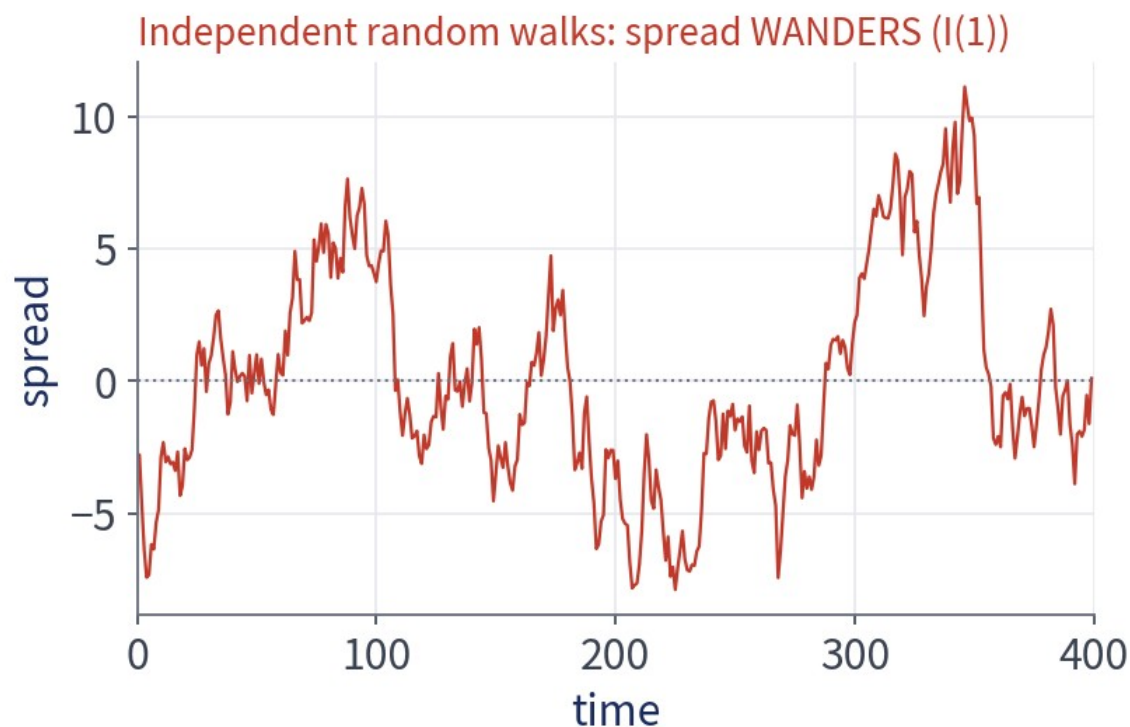
The cointegrated case

- The residual is stationary — $I(0)$.
- That stationarity is the certificate of a real relationship.
- So testing the residual is exactly how we tell them apart (Part 2).

Spurious vs Genuine

The residual is the tell. Independent random walks leave a spread that keeps wandering (left, $I(1)$); a truly cointegrated pair leaves a spread that mean-reverts (right, $I(0)$).

The tell: a genuine equilibrium spread is stationary; a spurious one is not



Cointegration vs Correlation

Related ideas, but not the same

Correlation

- Measures short-run co-movement of returns.
- Can be high one month and gone the next.
- Says nothing about a long-run tie in levels.

Cointegration

- A long-run equilibrium relationship in LEVELS.
- Deviations are temporary and get corrected.
- Two series can be cointegrated yet weakly correlated day-to-day.

The Cointegrating Vector

β is the hedge ratio in disguise

cointegrating vector $(1, -\beta)$: $z_t = Y_t - \beta X_t$ is the equilibrium error

What β means

- It is the weight that makes the combination stationary.
- The pair $(1, -\beta)$ is the cointegrating vector.
- z is the equilibrium error — the spread we watch.

The trading reading

- β is how many units of X hedge one unit of Y .
- It defines the market-neutral spread position.
- Estimate it once; it anchors the whole strategy.

Why Finance Loves Cointegration

A rare source of genuine mean reversion

Prices are close to random walks — hard to forecast. But a cointegrated SPREAD mean-reverts, giving a statistically grounded signal that survives where single-asset forecasting fails.

What it provides

- A mean-reverting quantity built from non-mean-reverting prices.
- A market-neutral position: long one, short the other.
- An economic anchor, not just a fitted pattern.

Where it shows up

- Relative-value and statistical-arbitrage strategies.
- Index-arbitrage and futures-basis trades.
- The whole discipline of pairs trading.

Examples in Markets

Where cointegration plausibly holds

Strong candidates

- Spot and futures on the same underlying.
- Dual-listed shares of the same company.
- Two firms in the same industry with shared drivers.

The economic logic

- An arbitrage or substitution force ties them together.
- When they diverge, that force pulls them back.
- No such force → no cointegration, however correlated.

The Common-Trend View

One shared random walk underneath

Cointegrated series can be seen as one shared stochastic trend plus stationary noise. The right β removes the common trend, and only the stationary parts remain — the spread.

The decomposition

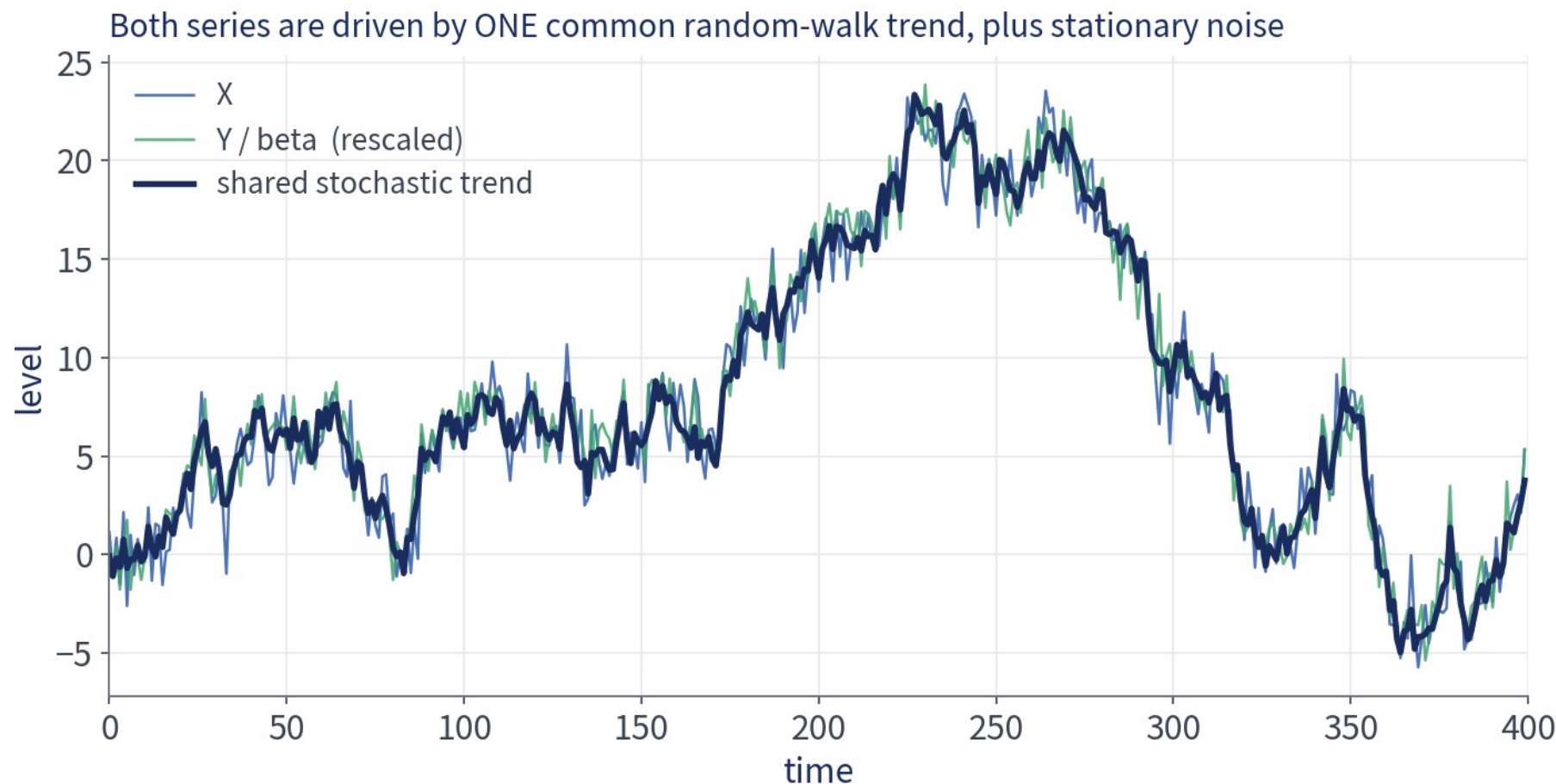
- Each series = a common random-walk trend + its own noise.
- The common trend is the source of the non-stationarity.
- It is SHARED, so it can be cancelled.

Why this clarifies it

- Cointegration = 'fewer trends than series'.
- Two series driven by one trend are cointegrated.
- Subtracting the shared trend leaves the stationary spread.

One Common Trend Underneath

Both series track a single shared random-walk trend (dark line), each adding only stationary noise. That shared trend is what the cointegrating combination cancels.



When Two Series Are NOT Cointegrated

No shared trend, no anchor

The signs

- Every linear combination still wanders — the spread is $I(1)$.
- There is no economic force binding them.
- High correlation can still occur, but there is no equilibrium.

Why it matters

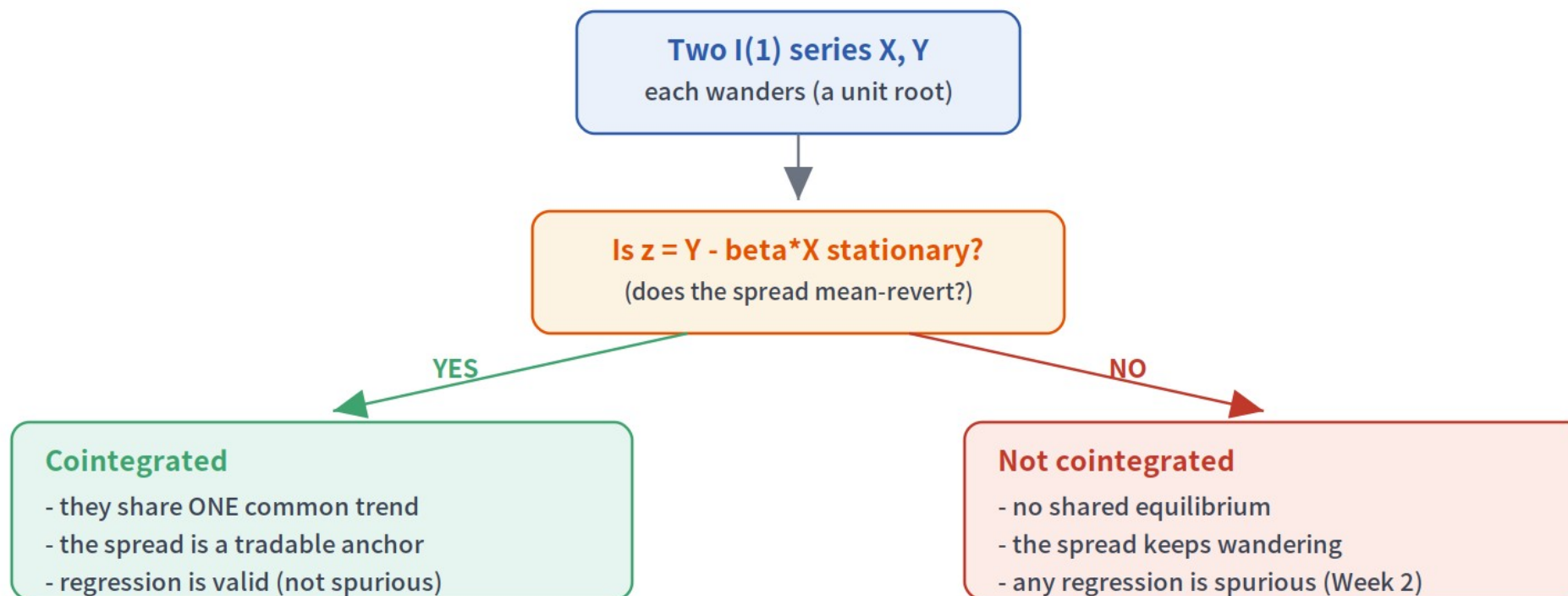
- Trading a non-cointegrated 'spread' is trading a random walk.
- It can drift away and never come back.
- Testing (Part 2) is what protects you from this.

Is a Pair Cointegrated?

Two $I(1)$ series are cointegrated only if some combination $Y - \beta X$ is stationary. If yes, they share a trend and give a tradable anchor; if no, any regression is spurious.

Is a Pair Cointegrated?

Two wandering series are cointegrated only if some linear combination of them stops wandering.



From Concept to Test to Trade

How the rest of the week unfolds

Part 2 — testing

- Engle-Granger: regress, then test the residual for stationarity.
- Johansen: a system test for several series at once.
- Read the cointegration rank — how many equilibria hold.

Part 3 — trading

- Error correction: the force that restores the equilibrium.
- The VECM and the speed of adjustment.
- Pairs trading Stage II, built on the cointegration spread.

Visualize a Candidate Pair

Direct Claude Code to see the anchor before testing

GOAL Take a plausible pair, estimate the spread, and see whether it looks stationary — the visual precursor to a formal test.

[INPUT] Prompt · Claude Code

Write and run `coint_intro.py` that:

1. downloads two related price series (e.g. two ETFs in the same sector, or a stock and its index) and confirms each is $I(1)$;
 2. regresses one on the other to estimate β , and forms the spread $z = Y - \beta X$;
 3. plots both prices together and the spread below, marking its mean;
 4. eyeballs whether the spread mean-reverts.
- Tell me: does this pair look cointegrated, and what is the estimated β (hedge ratio)?

How to work with it

- Both series should each be non-stationary ($I(1)$).
- A spread hugging its mean suggests cointegration.
- A spread that drifts away suggests none.
- Keep it; Part 2 turns eyeballing into a test.

Deliverable: `coint_intro.py` + a prices-and-spread plot with the estimated β .

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1	Two $I(1)$ prices have a stationary spread $Y - \beta X$. What are they?	Cointegrated
Q2	How does cointegration escape the spurious-regression trap?	Stationary residual
Q3	Can two series be highly correlated but not cointegrated?	Yes — different ideas

All three come from the definition, the stationary spread, and the correlation contrast.

Three Misconceptions to Avoid

Where cointegration intuition slips

Common myths

- "High correlation means cointegration."
- "Any two trending series can be paired."
- "A cointegrated spread can never drift far."

The reality

- Correlation is short-run; cointegration is a long-run levels tie.
- Only pairs sharing a common trend are cointegrated — most aren't.
- Spreads mean-revert on average, but can stray far first (Part 3).

Three Things to Carry Forward

Part 1 in one screen

1 Cointegration = a stationary combination of wanderers

Two $I(1)$ series are cointegrated if some combination $Y - \beta X$ is stationary. They share one common trend, and the spread is their long-run equilibrium error.

2 It cures spurious regression

A levels regression is genuine precisely when its residual is stationary. That stationary spread is the certificate — and testing it is how we confirm cointegration.

3 The spread is a tradable anchor

Prices barely mean-revert, but a cointegrated spread does. That is a rare, economically grounded signal — the foundation of pairs trading.

Next — Part 2: testing for cointegration with Engle-Granger and the Johansen system approach.